LAWDECK

Build Phases

Version 1.1 · The order of work, and what “done” means for each phase

Document control	
Version	1.1 (adds Stages A to C: before-code items, full checklist map, parallel work)
Date	20 October 2026
Based on	PRD v2.0; Engineering Rules v1.0; System Design v1.0
Builder	Founder, building with Claude Code

How to use this document

- Work through the phases in order. Each phase ends with an exit checklist; do not start the next phase until every box is ticked, or an exception is written down.
- Inside each phase, build one feature at a time using the seven-step workflow in the Engineering Rules (spec, plan, small steps, test like a student, checklists, commit).
- Durations are rough estimates for one person building part-time with an AI coding tool. They will move; the exit checklists will not.
- Costs per phase follow the start-up cost estimate of 20 October 2026.

Overview

#	Phase	Result	Rough time	Cost / month
0	Foundations	Accounts, repository, empty app online, Claude replies “OK”	3–5 days	$5–10
1	Shell, sign-in and data	Sign up, three-pane workspace, secure database	2–3 weeks	$5–10
2	Library and search	First course’s material searchable; reviewer approval	2–3 weeks	$5–15
3	Learn: ask and explain	Structured, sourced answers; model choice tested	3–4 weeks	$10–20
4	Practice	Questions, flashcards, marking, briefs	2–3 weeks	$10–20

#	Phase	Result	Rough time	Cost / month
5	Draft	Nigerian-format documents, preview, DOCX and PDF	3–4 weeks	$10–20
6	Uploads and sharing	Private uploads, sharing, account deletion	1–2 weeks	$10–20
7	Pre-pilot hardening	Payments, limits, backups, security, full library	2–3 weeks	$50–60
8	Closed pilot	10–20 students, then a larger cohort; real costs measured	4–6 weeks	$70–115

Total to a pilot-ready release: roughly 4–6 months part-time. Phases 2–5 start with one course (Civil Litigation) and widen to all five in Phase 7.

Stage A. Before any code (completed)

Everything that had to be decided before the first line of code is done. These are the foundations every phase builds on.

Item	Where it was settled	Status
Product requirements	PRD v2.0	Done
Engineering rules (how code is written)	Engineering Rules v1.0 and CLAUDE.md	Done
System design (the parts and how they connect)	System Design v1.0, sections 1–2	Done
Data modelling (the tables)	System Design, section 3	Done
Authorization (who can do what)	System Design, section 4	Done
Database permissions (row-level security plan)	System Design, section 4.3	Done
Environment variables (keys and settings list)	System Design, section 8.3	Done
Mobile-first rule	Engineering Rules M-01; phone design	Done
Error-handling pattern	Engineering Rules I-01 to I-03	Done
AI code security rules	Engineering Rules G-07, P-01, Appendix C	Done
Testing approach	Engineering Rules O-01 to O-06	Done
AI provider	Claude (Anthropic); Voyage AI for search	Done
Interface design	Lawdeck UI Design canvas (six boards)	Done
Brand: logo, colours, fonts	Brand board: monochrome logo, Newsreader, Inter, Outfit	Done (vector logo redraw pending)
Cost model	AI Cost Model spreadsheet and explainer	Done (finance review pending)
Legal position	Counsel’s advice of 20 October 2026, in PRD section 9.1	Done
Open items list	Open Items tab; PRD section 14	Kept up to date

Stage B. The full checklist: where each item happens

Each item on the engineering checklist is designed once, built in a specific phase, and checked again later. Nothing is left to “someday”.

Item	Designed (before code)	Built in	Checked again in
System design	Done: System Design	All phases follow it	End of every phase
Data modelling	Done: System Design §3	Phases 1–6, tables added as features need them	Every phase (permission tests)
Authorization	Done: System Design §4	Phase 1	Phase 7 security review
Database permissions	Done: row-level security plan	Phase 1, then every new table	Every phase; Phase 7 review
Environment variables	Done: §8.3 list	Phase 0	Phase 7 (separate staging and production keys)
Login	Rules C-01 to C-04	Phase 1	Phase 7 (login limits)
Mobile responsiveness	Rule M-01; phone design	Phase 1, then every screen	Every exit checklist
Error handling	Pattern I-01	Phase 1, then every feature	Phase 7 (error monitoring)
AI code security	Rules G-07, P-01, Appendix C	Phase 0 (review every change); Phase 3 (prompt-injection guard)	Phase 7 security review
Testing	Rules O-01 to O-06	Phase 1 (test tools); Phase 3 (AI evaluation)	Every exit checklist
Networking	Rules J-01 to J-03	Phase 0 (HTTPS); Phase 3 (streaming, retries)	Phase 7
Rate limiting	Rules K-01 to K-03	Phase 0 (spending cap); Phase 3 (usage logging, basic limits)	Phase 7 (plan limits, login limits)
Performance	Rules L-01 to L-03	Phase 1 (page budgets); Phase 3 (answer speed)	Phase 7 (load test)
Backups	Rules Q-01 to Q-03	Phase 2 (library source store); Phase 7 (daily backups)	Phase 7 (restore test)
Data breach check	Rules P-02 to P-05; Appendix D	Phase 0 (secret scanning)	Phase 7 (monitoring, response plan)

Stage C. Work that runs alongside the code

These are not coding tasks, but phases depend on them. Start them now so they are ready when the phase needs them.

Workstream	Who	Needed by
Collect library material: notes from 2020 onward, course materials and drafts for all five courses	Founder	Civil Litigation by Phase 2; all courses by Phase 7
Recruit legal reviewer(s) (lecturers or practitioners)	Founder and legal co-founder	Before Phase 2
Write evaluation questions and pushback cases	Reviewer	Phase 3
Write and approve rubrics	Reviewer	Phase 4
Review document templates and court formats	Reviewer	Phase 5
Finance review of the cost model	Finance and legal co-founder	Before Phase 3 model choice
Decide prices, free-tier limits and payment methods	Founders	Before Phase 7
Choose payment provider	Founder	Before Phase 7
Privacy Notice and Terms of Use	Legal co-founder	Before Phase 7
Redraw the logo as a vector (SVG) file	Founder or designer	Before Phase 1 design tokens
Partnership: NBA Young Lawyers Forum, then the Council of Legal Education	Legal co-founder	Demonstration after Phase 5; formal approach in Phase 8
Team members and hours (open item)	Founder	Ongoing; revisit before Phase 3

Phase 0. Foundations

Goal: Every account and tool is in place, an empty Lawdeck app is online, and Claude answers a test message.

Tasks

☐ Create the private GitHub repository lawdeck; turn on secret scanning and push protection (P-02).

☐ Add docs/ with the PRD, Engineering Rules, System Design and this document; add CLAUDE.md at the root (T-01).

☐ Create the Next.js app with TypeScript (strict) and Tailwind; set up the folder structure from the Engineering Rules (A-01, A-05).

☐ Create .env.example listing every variable; keep .env.local out of git (B-01 to B-03).

☐ Create a Supabase development project in a European region; switch on the pgvector extension.

☐ Connect the repository to Vercel; every pull request gets a preview link (R-02).

☐ Create the Anthropic developer account; load $5–10; set a $20 monthly limit and alerts (K-03).

☐ Create the Voyage AI account and key.

☐ Build the health check: a page that asks Claude “Reply with OK” through src/server/ai and shows the reply.

☐ Set up the branch habit: one feature per branch, merge by pull request (S-01 to S-03).

Rules that apply from this phase

Code structure (A), secrets and environment variables (B), dependencies (D-01), Git workflow (S), documentation (T), secret scanning (P-02), AI module (G-01).

Exit checklist (all must be true)

☐ The app is live on a Vercel address and shows a placeholder home page.

☐ The health check shows Claude’s reply; the API key is not visible anywhere in the browser.

☐ No secrets in the repository (secret scanning is on and clean).

☐ CLAUDE.md and the four documents are in the repository.

> **Note**
> Use the free plans of GitHub, Vercel and Supabase. The only payment is the first Claude credit.

Phase 1. Shell, sign-in and data foundation

Goal: A student can sign up, sign in and see the calm three-pane workspace on a phone and a computer, on top of a secure database.

Tasks

- ☐ Design tokens: colours, spacing, type, light and dark themes (M-03).
- ☐ Three-pane shell: left navigation, middle work area, right panel (closed by default); collapses correctly at 1024 px and 768 px (WS-01, M-01).
- ☐ Input-first home: greeting, composer, mode selector (Auto, Learn, Draft), course selector, attach button; no AI yet (WS-02).
- ☐ Sign-in with Supabase Auth: email and password with verification, Google sign-in, password reset (C-01 to C-03, AC-01).
- ☐ First migrations: profiles (with role), plans, projects, conversations, messages, courses, topics, usage_events, audit_log, all with row-level security (E-03, N-01 to N-03).
- ☐ Permission tests: two test students cannot see each other’s rows (E-06, O-02).
- ☐ Standard error pattern and loading, empty and error states (I-01, I-02).
- ☐ Test tools set up (Vitest, Playwright); tests run on every pull request (O-01, O-03, O-05).
- ☐ Dependency security check on pull requests (D-03).
- ☐ Projects: create, rename, delete; chats listed in the left pane (WS-05).

Rules that apply from this phase

Authentication (C), authorization and database permissions (E), validation (F-01, F-04), errors and logging (I-01 to I-03), mobile, accessibility and design (M), data modelling (N), testing (O-01 to O-03, O-05), dependencies (D-02, D-03).

Exit checklist (all must be true)

- ☐ Sign up, verify email, sign in, sign out and reset password all work.
- ☐ Workspace looks right and is usable at 360 px, 768 px, 1024 px and desktop widths; keyboard only works.
- ☐ Every table has row-level security; permission tests pass.
- ☐ Home page loads in under 3 seconds on a throttled 4G connection.

Phase 2. Library and search

Goal: The first course's material (Civil Litigation) is in the library, reviewed, and searchable by meaning and by exact words.

Tasks

- ☐ Migrations: kb_items, kb_passages with metadata, review status and embedding model (H-02).
- ☐ Admin upload screen for library files with their details; files stored in the private library bucket.
- ☐ Text extraction, OCR for scans, splitting into passages with headings and page numbers.
- ☐ Embeddings with Voyage AI; test the general and legal models on 30 Civil Litigation questions and record the choice.
- ☐ Hybrid search (meaning + keywords), filtered by course and approval status in the database.
- ☐ Reviewer console: list items, view passages, approve, correct, reject, withdraw; record reviewer and date (H-05, FB-02).
- ☐ Currency flag: items older than a later change in the law they discuss are marked for review (KB-02).
- ☐ Load and review the Civil Litigation material (notes from 2020 onward, course materials, drafts, rules of court).
- ☐ Keep the original source files in a separate versioned store (Q-03).

Rules that apply from this phase

Knowledge base (H), retrieval permissions (E-04), upload validation (F-02), no legal content in code (H-01).

Exit checklist (all must be true)

- ☐ Civil Litigation material is loaded, and every item shown to students is approved.
- ☐ For 30 test questions, the right passage appears in the top results at least 25 times.
- ☐ Unapproved and withdrawn items never appear in search results (tested).
- ☐ The embedding model choice is recorded with the test results.

Phase 3. Learn: ask and explain

Goal: A student asks a Civil Litigation question and gets a clear, structured, sourced answer that streams in, with honest limits.

Tasks

- ☐ Prompt files for explanations, with the answer structure and the behaviour rules (correct, hold position, accept valid correction) (G-02).
- ☐ Explain flow: retrieve, build prompt, stream answer, save message and sources (LN-01, LN-03).
- ☐ Sources panel: clicking an authority opens its passage in the right panel (WS-04).
- ☐ Citation check and 50-word quote limit on every answer (G-04, G-05).
- ☐ “Not enough support” response when passages cannot support an answer (LN-04).
- ☐ Simple and Exam-ready levels (LN-02); course hubs listing topics (LN-05).
- ☐ Auto mode routing with Haiku and the “Treated as” chip (WS-03).
- ☐ Usage logging per call, and per-user limits checked before each call (G-08, K-01 groundwork).
- ☐ Evaluation set v1 for Civil Litigation: at least 40 questions plus pushback cases, reviewed by a legal reviewer (O-04, O-06).
- ☐ Model quality test: run the evaluation with Haiku, Sonnet and Opus for explanations; choose by quality and cost; update the cost model.

Rules that apply from this phase

AI rules (G-01 to G-10), prompt injection protection (G-07), no personal data to Claude (G-06), networking and streaming (J), performance budgets (L-01, L-02), AI evaluation and pushback tests (O-04, O-06).

Exit checklist (all must be true)

- ☐ Zero fabricated authorities on the Civil Litigation evaluation set.
- ☐ At least 85% of evaluation answers rated “approved” by the reviewer.
- ☐ All pushback cases pass (corrects, holds, rejects fake authority, accepts valid correction).
- ☐ First words of an answer appear within 3 seconds (median).
- ☐ The explanation model is chosen and the cost per message is measured.

Phase 4. Practice

Goal: Students can practise with questions and flashcards, get their answers marked, and read case and statute briefs.

Tasks

- ☐ Migrations: practice_sets, practice_items, rubrics, attempts.
- ☐ Practice questions (multiple choice and scenario) by course and topic (LN-06).
- ☐ Flashcards with known and unknown tracking (LN-07).
- ☐ Rubrics written and approved for Civil Litigation topics (with the reviewer).
- ☐ Mark my answer: typed or uploaded answer, structured feedback in the right panel (LN-08).
- ☐ Case and statute briefs (LN-09).
- ☐ Marking agreement test: 50 answers marked by Lawdeck and by the reviewer.

Rules that apply from this phase

AI output validation (F-03), structured results, all earlier rules.

Exit checklist (all must be true)

- ☐ Lawdeck’s marking agrees with the reviewer on at least 80% of rubric points.
- ☐ Questions and flashcards cite only library material.
- ☐ All practice data passes the two-student permission tests.

Phase 5. Draft: Nigerian-format documents

Goal: A student asks for a Civil Litigation or Property Law document and gets it laid out correctly, previewed as pages, and exported to DOCX and PDF.

Tasks

- ☐ Migrations: forums, templates, documents, document_versions.
- ☐ Template and forum-profile schemas; validation before approval (H-03).
- ☐ One renderer for preview and DOCX (H-04); PDF converter running locally, then hosted.
- ☐ Draft flow: identify document, choose court or registry, intake questions, Opus fills narrative fields, placeholders for missing facts (DR-02 to DR-04).
- ☐ Automatic checks: parts, names, dates, defined terms (DR-06).
- ☐ Preview panel as A4 pages; edit in the panel or by instruction; versions (DR-05).
- ☐ “Why is it arranged this way?” explanations (DR-08) and related documents for a Lagos writ (DR-09).
- ☐ First templates, reviewed by a legal reviewer: writ of summons, statement of claim, motion on notice with affidavit, deed of assignment, tenancy agreement.
- ☐ Export tests: DOCX opens correctly in Word and Google Docs; PDF matches preview.

Rules that apply from this phase

Templates as data (H-03), one renderer (H-04), unreviewed content never shown (H-05).

Exit checklist (all must be true)

- ☐ Five templates approved by a reviewer, each with its rule source and review date.
- ☐ “Draft a writ of summons” produces a correct writ in the panel within 30 seconds.
- ☐ No generated document contains an invented party, date or property detail.
- ☐ Exported DOCX and PDF match the preview page for page.

Phase 6. Uploads, sharing and account controls

Goal: Students can upload their own study material privately, share documents view-only, and delete their account.

Tasks

- ☐ Upload notice shown before first upload (UP-06).
- ☐ Private uploads: storage folder rules, extraction, passages tied to owner (UP-01 to UP-05).
- ☐ Answers label “From your uploads” versus “From the Lawdeck library”.
- ☐ Sharing: view-only links, list of who has access, revoke (EX-04).
- ☐ Export notes, briefs and flashcards (EX-03).
- ☐ Delete account and data (AC-05); report a problem (FB-01); takedown request route (FB-03).

Rules that apply from this phase

Upload validation (F-02), storage rules (E-03), prompt injection protection for uploads (G-07).

Exit checklist (all must be true)

- ☐ A second student can never retrieve, see or be answered from another student’s upload (tested).
- ☐ Revoking a share stops access immediately.
- ☐ Deleting a file removes it from search within 5 minutes.

Phase 7. Pre-pilot hardening

Goal: Lawdeck is safe, affordable and complete enough for real students.

Tasks

- ☐ Move to Supabase Pro and Vercel Pro; create staging and production environments with separate keys (B-04, R-01).
- ☐ Payments with a Nigerian provider; plans and limits enforced (AC-03, AC-04, K-01).
- ☐ Login, sign-up and reset limits (C-04, K-02).
- ☐ Daily backups and a practice restore into staging (Q-01, Q-02).
- ☐ Error monitoring and unusual-access alerts (I-04, P-03).
- ☐ Breach response plan finalised (P-04, Appendix D).
- ☐ Security review of permissions, secrets and uploads (P-05).
- ☐ Load test for 500 concurrent users (L-03).
- ☐ Privacy Notice and Terms of Use reviewed by counsel; consent at sign-up.
- ☐ Library, rubrics and templates for all five courses loaded and approved.
- ☐ Full evaluation set (150+ questions, 30 drafting requests) passes all launch gates.

Rules that apply from this phase

All remaining “Pre-pilot” rules: C-04, I-04, K-01, K-02, L-03, P-03 to P-05, Q-01, Q-02.

Exit checklist (all must be true)

- ☐ Every PRD launch gate in section 11 is met.
- ☐ A restore from backup has been completed successfully.
- ☐ Security review findings are fixed.
- ☐ Prices and free-tier limits are set using the cost model and measured costs.

Phase 8. Closed pilot

Goal: Real Law School students use Lawdeck; we learn what works, what it costs and what they will pay.

Tasks

- ☐ Invite 10–20 students (start with resit candidates or recent finalists), then widen to a cohort.
- ☐ Weekly review of problem reports, evaluation results and costs.
- ☐ Measure: weekly active students, messages per student, cost per student, marking agreement, document use.
- ☐ Interview pilot students on value and price.
- ☐ Prepare the demonstration for the Council of Legal Education through the NBA Young Lawyers Forum.

Rules that apply from this phase

All rules apply. No feature merges without the full definition of done.

Exit checklist (all must be true)

- ☐ At least 40% of pilot students still active in week 4.
- ☐ At least 15% of pilot users willing to pay at the tested price.
- ☐ Real cost per student is known and within the target share of price.
- ☐ A go / revise / stop decision is recorded.

When each Engineering Rule takes effect

Phase	Rules
0 Foundations	A-01 to A-07, B-01 to B-03, B-05, D-01, G-01, P-01, P-02, R-02, S-01 to S-03, T-01, T-02
1 Shell, sign-in, data	C-01 to C-03, D-02, D-03, E-01 to E-03, E-05, E-06, F-01, F-04, I-01 to I-03, M-01 to M-03, N-01 to N-04, O-01 to O-03, O-05, T-03
2 Library and search	E-04, F-02, H-01 to H-03, H-05, Q-03
3 Learn	F-03, G-02 to G-10, J-01 to J-03, K-03, L-01, L-02, O-04, O-06
5 Draft	H-04
7 Pre-pilot	B-04, C-04, I-04, K-01, K-02, L-03, P-03 to P-05, Q-01, Q-02, R-01

Once a rule takes effect it applies to all later work. Rules labelled “Phase 1” in the Engineering Rules (meaning the first build stage) are spread across Phases 0–3 here; separate staging keys and environments (B-04, R-01) wait until Phase 7, when real users arrive.